

EDUCATION

University of Waterloo

B.A.Sc Computer Engineering

Waterloo, ON

Sep 2022 - Dec 2027

- Relevant courses: Reconfigurable Computing and ML Hardware Systems (Master's), Computer Architecture, Digital Hardware Systems

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, C/C++, Python, Vitis HLS, Assembly (ARMv7, RISC-V)

Protocols: AXI4/AXI4-Lite, AXI-Stream, PCIe, SPI, UART, Ethernet

Tools: Vivado (synthesis, STA, P&R, timing analyzer), Quartus Prime Pro, CocoTB, GTKWave, Git, Linux

Concepts: RTL design, HLS, timing closure, low-latency design, CDC, DMA, pipelining/retiming

EXPERIENCE

FPGA Engineer (Co-op)

ArchES Computing Systems

Toronto, ON

Sep 2026 - Present

- Integrated an AXI-Stream monitor into a large order-routing platform on a **Xilinx V80LL** and closed timing (**WNS -1.77 ns to +0.049 ns**), using the Vivado timing analyzer and block diagram viewer to trace critical paths and combinational loops; broke the loops with register slices and split a reset driving 30+ debug probes across two reset blocks (full builds took about a day)
- Designed the architecture of a packet capture tool and implemented it in **Vitis HLS**; it records packets from Ethernet RX or an exchange simulator and sends them to the host over an **HMQ ring** for PCAP generation, to speed up error reproduction across **10+ platforms**

FPGA Research Assistant

University of Waterloo - Under Prof. Mina Arashloo

Waterloo, ON

Jan 2026 - Aug 2026

- Built components of a protocol-agnostic FPGA **transport-layer acceleration** architecture on **AMD Alveo U250**, enabling hardware offload of multiple network protocols (TCP, RoCEv2)
- Implemented **PCIe/QDMA** host ↔ FPGA communication using AXI-Stream in the OpenNIC 250 MHz user logic, allowing applications to drive FPGA transport logic
- Built **buffering, backpressure, and request-framing** logic to reliably inject host requests into a high-throughput FPGA networking pipeline

Digital Design Team Lead - Ethernet Packet Parser

UW ASIC Design Team

Waterloo, ON

Aug 2025 - Present

- Leading a team of **15 members** on the architecture and RTL of a high-throughput **Ethernet packet parser**: a multi-layer pipeline for Ethernet/IP/TCP header extraction with configurable match-action rules, targeting **10G line rate**
- Led RTL and **floorplanning** of a cryptography accelerator, closing timing at **200 MHz** with **65% FPGA resource utilization**

TECHNICAL PROJECTS

GEMM on AMD EPYC and Tenstorrent Wormhole C, AVX-512, OpenMP, TT-Metal

- Wrote a **BF16 GEMM kernel** with **AVX-512BF16** intrinsics that reaches **2,879 GFLOPS (93% of theoretical peak)**, **44% faster** than AMD's AOCL library (1,994 GFLOPS) and **ranked 1st** on the course leaderboard (ECE 493/720)
- Optimized with SIMD-interleaved B matrix pre-packing, 6-row accumulator unrolling for register reuse, a 4-way unrolled k-loop, and **OpenMP** across N-tile slices
- Built a **multi-core GEMM** for the Tenstorrent Wormhole accelerator in TT-Metal (reader, compute, and writer kernels over the Tensix NoC mesh), reaching **2,570 GFLOPS** at a 6x6 core grid, a **24x** speedup over one core

BERT Encoder on AMD Versal Hard-NoC (FPGA Simulation) SystemVerilog, Tcl, AMD Versal, NoC, AXI

- Architected a **transformer encoder datapath** on AMD Versal with 4-head self-attention over parallel AXI-Stream + DMA pipelines; **1.5 GB/s NoC bandwidth** at **300 MHz** measured in RTL simulation
- Cut memory bandwidth **4x** with INT8 quantization; parameterized tokens, embedding size, and head count, with systolic tiling and AXI burst alignment constraints

High-Frequency Adders on Intel Agilex-5 FPGA SystemVerilog, Quartus Prime Pro, TimeQuest, Bash

- Optimized a 2048-bit adder for **41% higher frequency** (600→848 MHz) and **25% less area**; closed timing with TimeQuest STA using retiming and pipelining
- Automated a sweep of **11 configurations** with Bash scripts that drive Quartus synthesis and parse timing reports to pick the best frequency-area tradeoff